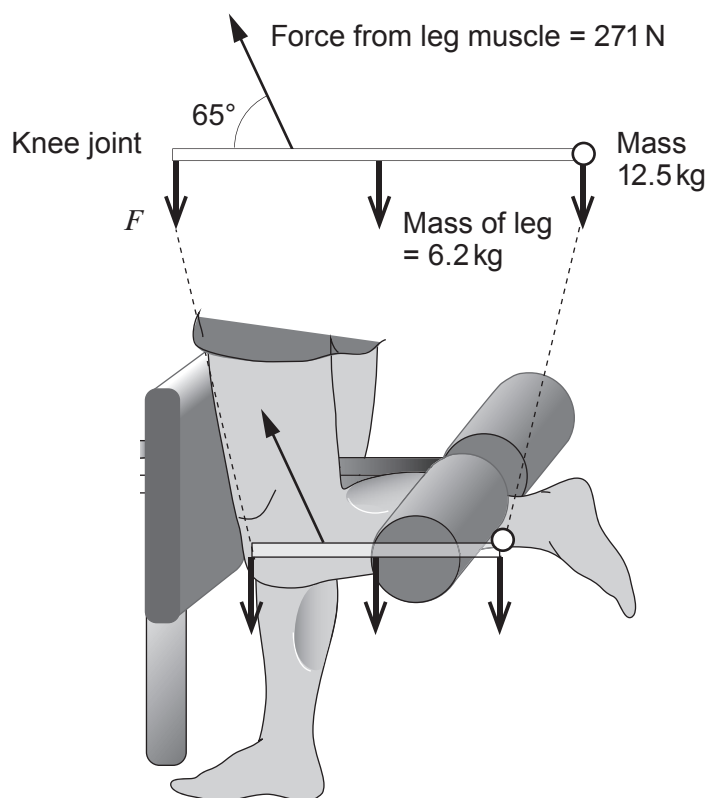


Option C – The Physics of Sports

8. The following simplified diagram is of the lower leg of an athlete holding a mass of 12.5 kg in equilibrium. The leg pivots around the knee joint and the effective mass of the lower leg is 6.2 kg. The force, F , shown is the vertical component of the force exerted on the lower leg by the knee.



(a) Calculate the force, F .

[3]

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- (b) (i) A football player must kick a football a distance of 43.6 m for it to be a successful kick. Determine whether or not the kick will be successful if it is kicked with a speed of 18 ms^{-1} at an angle of 50° to the horizontal. *Ignore the effects of air resistance.* [4]

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- (ii) The football has a mass of 0.45 kg and spins at a rate of 8.8 Hz. It has a moment of inertia of 0.0041 kg m^2 .

- I. Explain what is meant by the moment of inertia. [2]

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- II. Calculate the angular momentum of the football. [2]

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- III. Determine the **total** kinetic energy of the ball **at its greatest height** (*remember to ignore the effects of air resistance*). [4]

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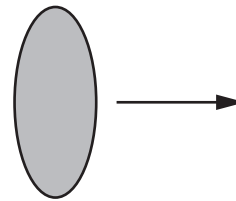
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- (c) Due to the shape of a rugby ball, its cross-sectional area can be varied depending on its orientation and motion. The minimum and maximum cross-sectional areas are 43.2 cm^2 and 217.7 cm^2 respectively and are shown below.



Minimum cross-sectional
area = 43.2 cm^2



Maximum cross-sectional
area = 217.7 cm^2

- (i) Explain clearly by what factor the drag force changes for these two cross-sectional areas. [3]

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- (ii) If the rugby ball had been kicked in a region where the density of the air is lower, explain how your answer to part (c)(i) would change, if at all. [2]

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